

Windows + WE Home Server Setup Guide

Using a powerful Windows PC at home as a server for your Flutter app and AI backend.

Overview

This guide explains how to use a strong Windows PC at home as a backend server for your Flutter application. The backend runs FastAPI with AI models, accessible securely over the internet via WE (Telecom Egypt).

Step 1 — Prepare the Windows Server

Install Python 3.10+, copy your backend project folder, create a virtual environment, and install required packages. Run FastAPI locally and verify it works on localhost.

Step 2 — Run the Backend Server

Start FastAPI using uvicorn with host 0.0.0.0 on port 8000. Confirm the health endpoint responds correctly.

Step 3 — Assign Static Local IP

Configure a static LAN IP for the server PC (e.g., 192.168.1.50) to ensure port forwarding always points to the correct machine.

Step 4 — Port Forwarding on WE Router

Log into the WE router (192.168.1.1) and add a Port Forwarding rule to map external port 8000 to internal port 8000 on the server PC.

Step 5 — Access from Outside the Home Network

Use your public IP to test access from mobile data. If successful, the server is reachable from the internet.

Step 6 — Handle Changing Public IP (DDNS)

Register a free DDNS domain (DuckDNS recommended) to keep a constant domain name even if your public IP changes.

Step 7 — Enable HTTPS (Recommended)

Install Caddy on Windows and configure it as a reverse proxy to provide automatic HTTPS for secure communication.

Step 8 — Connect Flutter App

Update the Flutter app baseUrl to point to your DDNS domain so the app can access the backend from anywhere.

Security Notes

Always protect your API with authentication, enable HTTPS, and consider rate limiting to prevent abuse.

Architecture Diagram

You (outside home) use the app → requests go to your home server

